BRYAN CHAN

Lead AI Platform Engineer

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📍 Toronto, Ontario, Canada

PROFESSIONAL SUMMARY

Full-stack TypeScript engineer with 8+ years building enterprise logistics and warehouse management platforms, now gone deep on agentic AI. At GoBolt, builds the AI agents that review PRs, maintain docs, and run releases — then uses them to ship faster than a team. Specializes in NestJS backend architecture, React frontends, and cloud-native deployments.

Core Expertise: TypeScript, NestJS, React, PostgreSQL, Kubernetes, Agentic AI & LLM Integration

TECHNICAL SKILLS

Frontend: React 18, TypeScript, Recoil, React Query, Vite, Ant Design, styled-components, Storybook, Playwright, Vitest

Backend: NestJS, Node.js, Express, TypeORM, PostgreSQL, Socket.io, Background Jobs, Distributed Locking

Cloud & DevOps: Kubernetes, Docker, ArgoCD, Helm, AWS (Elastic Beanstalk, Lambda), SOPS/KMS, GitOps, DataDog APM

Architecture: Domain-Driven Design, Microservices, Event-Driven Architecture, Message Queues, State Machines

Testing: Playwright, Vitest, Jest, FactoryBot, ESLint, Prettier, TypeScript Strict Mode

Tools: Git, LaunchDarkly, Sentry, Swagger/OpenAPI, DataDog APM

PROFESSIONAL EXPERIENCE

Lead AI Platform Engineer

September 2026 - Present

GoBolt (formerly Bolt Logistics) | Toronto, Ontario, Canada

- Built and operate a **fleet of production AI agents** on shared internal tooling — PR review and approval, documentation maintenance, release automation
- Released the first **WMS MCP server**, extended to the merchant domain, read-only tools exposing warehouse locations, floor stock, product master data, ledger balances, inbound receiving and outbound work orders, plus merchant domain data, behind OAuth sign-in and per-user scoping
- Shipped the first **fully automated agentic delivery flow**, where a product manager takes a feature from request to shipped end to end
- Shipped the company's first LLM-powered feature in the warehouse management system, with a reusable multi-provider LLM integration layer
- Created agent-maintained **knowledge bases for the warehouse and merchant domains** that keep themselves up to date from releases and team discussions
- Made agentic AI the core delivery workflow — AI-driven testing, releases, and code review
- Built the team's **end-to-end testing engine** for AI-generated code: one CLI brings up the warehouse, merchant and handheld apps against seeded local data, AI agents write and run the suites, and every release gets evidence it will not break production

- Led engineering-wide enablement on AI-assisted development: presented GoBolt's AI development workflow and trained engineers on Claude Code, agent workflows, and internal tooling

Technologies: *TypeScript, NestJS, Claude API, Gemini/Vertex AI, Anthropic SDK, MCP, Playwright, Kubernetes, DataDog*

Lead Software Engineer

October 2021 - September 2026 (5 years)

GoBolt (formerly Bolt Logistics) | Toronto, Ontario, Canada

Enterprise Warehouse Management System Architecture

- Architected and maintain enterprise warehouse management backend with **136+ modules** using NestJS, TypeORM, and PostgreSQL, processing thousands of daily warehouse operations across receiving, picking, packing, returns, and carrier transfer workflows
- Designed domain-driven architecture with standardized module patterns, custom decorators for authentication/permissions, and comprehensive Swagger API documentation
- Implemented **Generic Inflator Pattern** for background processing with distributed locking, supporting picksheet inflation, work orders, and asynchronous task execution with automatic retry and stale item cleanup
- Built **Signal Emitter Pattern** with two-table design for reliable HTTP request queuing to Core API with automatic retries and comprehensive event tracking
- Migrated from Redis-based distributed locking to PostgreSQL-based Generic Lock Service, eliminating Redis dependency and simplifying infrastructure
- Led a cross-team data-integrity project unifying product identity across systems, rolled out to production with zero incidents

Frontend Development & Real-Time Systems

- Led development of React 18 warehouse management frontend with **130+ routes** across 35+ feature areas using TypeScript, Vite, Ant Design, and Recoil state management
- Architected scanner-first mobile-optimized interface for warehouse floor operations with auto-focus input handling and Command+K fuzzy search navigation
- Implemented WebSocket integration for real-time packing operations with live status updates
- Built multi-step picking/packing wizards, ASN processing flows, inventory management, and special projects interfaces
- Designed state management architecture using Recoil for global state, React Query for server-side caching, and URL params for shareable flow state

Cloud Infrastructure & DevOps

- Managed Kubernetes deployments across **8 environments** (prod, preprod, meta, air, escape, mini, tide, tri) using Helm charts and ArgoCD GitOps workflows
- Implemented secrets management with SOPS encryption and KMS integration for secure configuration across environments
- Built Docker containerization strategies with multi-stage builds, health check endpoints, and optimized caching
- Integrated DataDog APM with custom decorators for performance monitoring and error tracking across all services

Testing & Code Quality

- Established testing patterns: Unit tests with FactoryBot, integration tests with database fixtures, E2E tests with Playwright
- Implemented Storybook for component documentation and visual regression testing
- Maintained test coverage with Vitest and Jest across all services
- Enforced code quality standards with ESLint, Prettier, and TypeScript strict mode

Technologies: *TypeScript, NestJS, React 18, PostgreSQL, Socket.io, Kubernetes, Docker, ArgoCD, Helm, TypeORM, Recoil, React Query, Ant Design, Vite, Playwright, Vitest, DataDog, LaunchDarkly, SOPS*

Full Stack Engineer, Senior Software Developer, Team Lead

November 2020 - October 2021 (1 year)

Bolt Logistics | Toronto, Ontario, Canada

- Built a new app with NestJS (TypeScript backend framework) for warehouse inventory management system
- Deployed the app on Amazon Elastic Beanstalk with proper health checks and monitoring
- Designed database architecture and entity relationships for warehouse operations
- Built a new ReactJS frontend interface for warehouse system
- Built serverless AWS Lambda functions to query secondary database
- Handled concurrency issues with distributed locks
- Built an access management app allowing team leads to grant feature access to team members

Lead Frontend Developer

December 2018 - November 2021 (3 years)

Bolt Logistics | Toronto, Canada Area

- Introduced React hooks and formalized them into the codebase
- Introduced TypeScript into the codebase
- Introduced styled-components and replaced SASS from the codebase
- Introduced npm packages for both web (React) and mobile (React Native) components
- Introduced react-query into the codebase
- Built a React app for vehicle routing system with Google Maps API
- Introduced RecoilJS into the system and replaced Redux, reducing 60% of boilerplate code

Frontend Developer

January 2018 - December 2018 (1 year)

Bolt Logistics | Toronto, Ontario, Canada

- Introduced Redux into main site and solved the prop drilling problem
- Implemented progressive data loading on management tool to improve performance
- Introduced SASS into the codebase
- Built and deployed first React Native app for operations team
- Built business site from scratch with React

Frontend Developer (Intern)

November 2017 - December 2017 (2 months)

Bolt Logistics | Toronto, Ontario, Canada

- Updated site footer on main site
- Implemented Stripe Element with React on the site

PERSONAL PROJECTS

BryanStudio.dev - Personal Portfolio Website

2025

- Built modern portfolio website using React 19, TypeScript, Vite, and Tailwind CSS
- Implemented Framer Motion animations and glassmorphism design aesthetics
- Deployed on Railway with Docker containerization and Express server
- Features multilingual support and responsive mobile-first design

Technologies: *React 19, TypeScript, Vite, Tailwind CSS, Framer Motion, Docker, Railway*

AI Career Chat Agent

2025

- Developed AI-powered chat agent using OpenAI GPT-4 with custom function calling and structured outputs
- Implemented PDF profile parsing for context injection and personalized career guidance
- Built modern ChatGPT-style interface with real-time streaming responses
- Integrated push notifications via Pushover and deployed on Railway

Technologies: *TypeScript, OpenAI GPT-4, Express, React, PDF parsing, Railway*

PetCrew - iOS Pet Management App

2025

- Developed privacy-first iOS app for pet care management using SwiftUI
- Implemented local-first architecture with household sharing capabilities
- Designed with zero analytics and telemetry for maximum user privacy
- Published to App Store with full legal compliance (Privacy Policy, Terms, Support)

Technologies: *SwiftUI, iOS, Local-first architecture*

EDUCATION

Bachelor of Applied Science (BASc), Computer Engineering

University of Toronto

KEY ACHIEVEMENTS

- Built and operate a fleet of production AI agents — PR review, docs maintenance, release automation
- Shipped the company's first LLM-powered feature with a reusable multi-provider integration layer
- Trained the engineering team on AI-assisted development and Claude Code adoption
- Architected enterprise warehouse management backend with 136+ modules across 10 warehouses
- Led React 18 frontend development with 130+ routes and real-time features
- Managed Kubernetes/ArgoCD deployments across 8 environments
- Designed distributed systems patterns (Generic Inflator, Signal Emitter, Distributed Locking)
- 8+ years at GoBolt showing loyalty, growth, and technical leadership

LANGUAGES

English (Fluent) • Mandarin Chinese (Fluent) • Cantonese (Fluent)